Anomaly Detection for Hyperspectral Imagery via Tensor Low-Rank Approximation With Multiple Subspace Learning

Xu He, Jing Wu, Qiang Ling, Zhaoxu Li, Zaiping Lin, and Shilin Zhou

***Abstract*— Hyperspectral anomaly detection (HAD) is regarded as an indispensable, pivotal technology in remote sensing and Earth science domains. Nevertheless, most existing detection approaches for anomaly targets flatten 3-D hyperspectral images (HSIs) with spatial and spectral information into 2-D spectral vector data, which virtually breaks up the internal spatial structure in HSIs and degenerates the detection performance. To this end, we directly consider the HSI data cube as a 3-D tensor and develop a novel tensor low-rank approximation (TLRA) detection algorithm to separate the sparse anomalous component from the background with low-rank characteristics. Then, in light of the multisubspace structure in heterogeneous backgrounds, we utilize multiple subspace learning (MSL) theory to encode the background tensor with a coefficient tensor and corresponding dictionary tensor. In addition, considering that different singular values indicate different information quantities and should be penalized to different extents, we introduce a tighter tensor rank surrogate named the ϵ-shrinkage tensor nuclear norm (ϵ-TNN) to recover the low-rank component more accurately. Meanwhile, concerning the sparse anomaly target, the $l_{2,1}$ constraint is incorporated to represent the group sparsity of the abnormal component. Finally, an effective iterative optimization algorithm based on the alternating direction method of multipliers (ADMM) is devised to solve the proposed TLRA-MSL model. We conduct extensive experiments on six hyperspectral datasets to prove the effectiveness and robustness of our method. The experimental results illustrate that better detection performance is obtained using the proposed model compared with other state-of-the-art algorithms.**

***Index Terms*— ϵ-shrinkage tensor nuclear norm (ϵ-TNN), anomaly detection, hyperspectral images (HSIs), multiple subspace learning (MSL), tensor low-rank approximation (TLRA).**

NOMENCLATURE

$\mathcal{A}/\boldsymbol{A}/\mathbf{a}/a$	Tensor/matrix/vector/scalar.
$\mathcal{A}(i,:,:)/\mathcal{A}(:,i,:)/$ $\mathcal{A}(:,:,i)$ or $\mathcal{A}^{(i)}$	ith horizontal slice/lateral slice/frontal slice of tensor $\mathcal{A}$.
$\mathcal{A}(:,i,j)/\mathcal{A}(i,:,j)/$ $\mathcal{A}(i,j,:)$	Mode-1 fiber/mode-2 fiber/mode-3 fiber (tube) of tensor $\mathcal{A}$.
$\mathcal{A}_{ijk}$ or a_{ijk}	(i,j,k)th index value of tensor $\mathcal{A}$.
$\boldsymbol{I}_n$	Identity matrix $\boldsymbol{I}_n \in \mathbb{R}^{n\times n}$.
$\mathcal{I}$	First frontal slice and other frontal slices of the identity tensor $\mathcal{I} \in \mathbb{R}^{n\times n\times n_3}$ are $\boldsymbol{I}_n$ and zeros matrix.
$\boldsymbol{A}^T$	Transpose of matrix $\boldsymbol{A}$.
$\mathcal{A}^*$	Conjugate transpose of tensor $\mathcal{A}$, see Definition 1.
$\boldsymbol{A}\cdot\boldsymbol{B}$	Matmul product between matrices $\boldsymbol{A}$ and $\boldsymbol{B}$.
$\mathcal{A} * \mathcal{B}$	Tensor product between tensor $\mathcal{A}$ and $\mathcal{B}$, which is defined in Definition 2.
$\|\mathbf{a}\|_2$	l_2 norm of vector $\mathbf{a}$, which is defined as $\sqrt{\sum_i a_i^2}$.
$\|\mathcal{A}\|_\infty$	Infinity norm of $\mathcal{A}$, which is defined as $\max_{ijk}\lvert a_{ijk}\rvert$.
$\|\mathcal{A}\|_{\mathrm{F}}$	Frobenius norm of $\mathcal{A}$, which is defined as $\sqrt{\sum_{ijk}\lvert a_{ijk}\rvert^2}$.
$\sigma_i(\boldsymbol{A})$	Singular values of matrix $\boldsymbol{A}$.
$\|\boldsymbol{A}\|_* = \sum_i \sigma_i(\boldsymbol{A})$	Nuclear norm of matrix $\boldsymbol{A}$.
$\overline{\mathcal{A}} = \mathrm{fft}\ (\mathcal{A}, [\ ], 3)/$ $\mathcal{A} = \mathrm{ifft}\ (\overline{\mathcal{A}}, [\ ], 3)$	Fast Fourier transform of $\mathcal{A}$ and inverse fast Fourier transform of $\overline{\mathcal{A}}$ along the third dimension.

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I. INTRODUCTION

THE hyperspectral images (HSIs) collected from airborne or satellite imaging sensors can effectively record the spectral characteristics of the observed objects in the same shooting scene. Meanwhile, there are discriminative differences between the spectral radiation characteristic curves of heterogeneous targets. Therefore, there is a sufficient scientific theoretical basis to distinguish different categories of objects through hyperspectral imagery. At present, mainstream research hotspots of HSI processing mainly consist of data reduction [1], [2], clustering [3], [4], unmixing [5], [6], classification [7], [8], denoising [9], [10], target detection [11], [12], [13], [14], [15], and so on. More precisely, the target detection task in hyperspectral imagery can be further classified into target matching detection and anomaly detection. Specifically, target matching detection can extract the actual object attributes from the ground target information in HSIs

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by analyzing and comparing the existing samples with the HSI. In contrast, in hyperspectral anomaly detection (HAD), the spectral prior information of the ground object is not provided. The abnormal targets that are significantly different from most of the surrounding background areas in HSIs are directly detected. For example, rare mineral distribution sites, abnormally developed crops, vehicles running on the road, or artificial artifacts in the natural background can be taken as detection objects in HAD. Thus, it has broad application prospects in geological prospecting, smart agriculture, resource exploration, and military reconnaissance [16], [17]. Nevertheless, limited by finite spatial resolutions and long imaging distances of the imaging sensor, the anomalies in HSIs usually appear as small areas embedded in a surrounding background with a low probability. Furthermore, affected by a sophisticated imaging environment, the heterogeneous backgrounds of HSIs also increase the difficulty of the HAD task. Therefore, distinguishing anomalies from complex backgrounds in HSIs has become a momentous and challenging issue that remains to be investigated. Recently, a growing number of researchers have paid attention to the HAD issue and developed various effective HAD algorithms. To the best of our knowledge, off-the-shelf HAD methods can be divided roughly into four representative categories: 1) statistical theory-based model; 2) distance measurement-based model; 3) representation learning-based model; and 4) low-rank and sparse factorization (LRSF)-based model.

A. Related Work

1) Statistical Theory-Based Model: This type of model was first developed in the field of HAD. The key principle of this method is to utilize global or local statistical models to estimate the distribution of background pixels and regard the pixels that are far away from the background distribution as abnormal targets. Statistical theory-based models usually make the strong prior assumption that the background pixels in HSIs obey some known mathematical statistical distribution model. For example, in the classical Reed–Xiaoli (RX) [18] algorithm, it is posited that the background pixels fall within the distribution of multivariate Gaussian. Specifically, the global-RX detector (GRXD) draws upon all pixels of the HSI to estimate the background distribution by calculating the background mean and covariance. Then, the anomaly degrees of test pixels can be measured by calculating the corresponding Mahalanobis distance. In contrast, in the local-RX detector (LRXD), a local window is devised, and the background covariance matrix is calculated by computing only the configured local window pixels. Based on the classical RX model, several improved models have been developed to address the problems of a single background statistical distribution and background contamination from anomalies or noisy pixels in the RX algorithm. For instance, Guo et al. [19] proposed the weighted-RX detector (WRXD) to enhance the adaptive parameter selectivity of the background statistics. Schaum [20] developed the subspace RX (SSRX) detector to eliminate background statistics contamination from anomalies or noisy pixels. Kwon and Nasrabadi [21] devised the kernel RX detector (KRXD), which projects hyperspectral data into a higher dimensional space through a nonlinear kernel transformation to enhance spectrum separability. In addition, Zhou et al. [22] effectively improved the computational efficiency of the original KRXD by comprehensively utilizing global and local information in HSI background statistics.

2) Distance Measurement-Based Model: In this model, pixels are grouped according to their distance measurement, and the pixels that deviate from the cluster center are labeled as anomalous pixels. According to different packet modes, the distance measurement-based model can be further divided into three categories: 1) cluster-based models; 2) support vector machine (SVM)-based models; and 3) graph theory-based models. Representative approaches of the cluster-based model include the k-means and Gaussian mixture model (GMM)-based HAD method proposed by Carlotto [23] and the density peak cluster-based HAD algorithm proposed by Tu et al. [24]. Based on the principle of the SVM mechanism, Banerjee et al. [25] proposed the classical support vector data description (SVDD) method to represent the local background of HSIs. Gurram and Kwon [26] presented a novel SVDD variant, in which the Gaussian radial basis function (RBF) kernel is applied to construct a compact hypersphere with minimum volume. The graph theory-based HAD method is used to calculate the similarity between any two pixels. If the similarity exceeds the given threshold, the corresponding pixels are connected to the edges. The majority of corresponding pixels are classified as background to build a background set. For example, Yuan et al. [27] applied a graph theory to the HAD task and improved the detection performance. Similarly, the kernel isolation forest-based detector (KIFD) [28] and its promoted version [29] were used to leverage the classical isolation forest (iForest) algorithm to accurately detect the anomalies in HSIs.

3) Representation Learning-Based Model: This model leverages the representation coefficient φ and specific dictionary D to reconstruct all the HSI pixels $H \approx D\varphi$. Meanwhile, the normal pixels, i.e., the background pixels, can be represented by several atoms in an overcomplete dictionary, while abnormal pixels cannot or with a relatively large reconstruction error. The larger the reconstruction error is, the more likely that the pixel to be tested is abnormal. According to different regulation constraints of φ, the representation learning-based model can be divided into sparse representation (SR)-based models [30], [31] and collaborative representation (CR)-based models [32], [33]. The SR-based model requires as few background atoms as possible in the over-complete dictionary to obtain a better linear representation. For example, in [31], the sum-to-one and nonnegativity constraints were imposed on the representation coefficient φ to strengthen the physical meaning in the SR-based model. Different from the SR-based model, the CR-based model represents each background pixel with all atoms in the background dictionary and regards the representation residual as the anomaly degree criterion. For instance, Su et al. [33] introduced a principal component analysis (PCA) into the CR-based background dictionary construction and effectively removed anomalous pixel contamination. Besides that, Vafadar and Ghassemian [34] proposed a

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modified-CR algorithm that utilizes the Gaussian distribution model to remove outliers within a sliding dual window.

4) Low-Rank and Sparse Factorization-Based Model: The core mechanism of this model is to transform the HAD problem into an LRSF problem, which fully considers the sparsity of anomalies and the low-rank attributes of the background. This type of detection model usually unfolds the 3-D hyperspectral data cube along with a spatial mode or spectral mode and leverages a matrix-based optimization method to separate the anomalies and background. Chen et al. [35] proposed the most representative robust PCA (RPCA) model, which achieves excellent detection performance. Inspired by the basic framework of RPCA [36], [37], [38], more improved detection algorithms were proposed. For example, Farrell and Mersereau [39] and Billor et al. [40] boosted the performance of RPCA model by exploiting a better representation of sparse anomalies. Sun et al. [41] introduced the Euclidean distance to describe the anomaly degree of the sparse component. In addition, to obtain better prior knowledge of the background, Zhang et al. [42] applied the Mahalanobis distance to estimate the background part with low-rank attribute. Cheng and Wang [43] combined the total variation (TV) and graph constraints to better exploit the spatial relationship information. Similarly, Feng et al. [44] incorporated the local spatial and TV regularization to extract the local spatial structure of the background. Cheng and Wang [45] characterized both background and anomaly components with a union dictionary. Niu and Wang [46] designed a learned background dictionary construction method to obtain better background representation. Furthermore, Li et al. [47] proposed a Manhattan distance-based detector and mixture of Gaussian (MoG)-based sparse modeling LRSF method.

B. Motivation

Although the abovementioned LRSF-based methods have achieved promising detection performance, it is also nonnegligible that the HSIs are intrinsically 3-D tensors with the union of two spatial dimensions and one spectral dimension. Consequently, many LRSF models have been proposed successively based on the 3-D tensor structure property of HSIs [48], [49], [50], [51], [52], [53], [54], [55], [56]. For example, Wang et al. [51] employed the CANDECOMP/PARAFAC (CP) decomposition to factorize the background tensor into the sum of several rank-1 tensors to reconstruct the background component. Li et al. [52] exploited a tensor Tucker decomposition-based HAD, where the decomposed core tensor was utilized to represent the background part. Furthermore, based on the Tucker decomposition, Zhang and Wen [53] implemented an adaptive Tucker rank calculation method to obtain the major principal component numbers of the core tensor. However, the Tucker rank is actually composed of the rank of the tensor mode-n unfolding matrix, which destroys the intrinsic structure of the tensor due to the matricization operation. The tensor tubal rank based on the tensor singular value decomposition (t-SVD) can effectively address the problems of matricization in the Tucker rank. With that in mind, Xu et al. [54] designed an efficient t-SVD-based anomaly detection algorithm. As a convex approximation of the nonconvex tensor tubal rank, the tensor nuclear norm (TNN) was utilized in [54] to estimate the low-rank part. However, TNN will bring about overshrinkage problem, which indicates that all singular values of tensors will be penalized equally. In general, a larger singular value implies a higher quality of information and should be penalized by a larger weight. To address this issue, Wang et al. [55] introduced the weighted TNN (WTNN). Nevertheless, the employment of the WTNN only a trifle retards issue of overshrinkage. Meanwhile, in a majority of the above tensor-based HAD methods, the intrinsic multisubspace structure of the background is neglected, and it is assumed that the background in an HSI is derived from a unitary low-rank subspace. Therefore, performance degradation will occur when using these methods for real scenes with heterogeneous backgrounds. To solve the above problems, a novel tensor low-rank approximation with multiple subspace learning (TLRA-MSL) model for HAD is proposed in this article. First, the 3-D hyperspectral data will be preprocessed using an orthogonal transformation to reduce the computation cost. The preprocessed HSI tensor is chopped up into an anomalous tensor and a background tensor. Meanwhile, considering the intrinsic multisubspace property in the background, we implement the multisubspace learning strategy, which utilizes a dictionary tensor and the corresponding coefficient tensor to obtain the linear representation of the heterogeneous background. In addition, we incorporate a tighter convex surrogate of the tensor tubal rank named the ϵ-shrinkage TNN (ϵ-TNN) to depict the low-rank component. Compared with the TNN, ϵ-TNN can be used to address the problem of overshrinkage by imposing different penalties for different singular values. Moreover, the $l_{2,1}$ norm is employed to characterize the group sparsity of the anomalous component. In conclusion, the primary contributions of this work are summarized as follows.

1) We develop a tensor low-rank approximation (TLRA) model based on the intrinsic 3-D hyperspectral tensor data. The multisubspace learning strategy is exploited to reconstruct the background component in view of the multisubspace background structure property. Moreover, the group sparse constraint is incorporated into our method to better dissociate the anomalous component from the original hyperspectral tensor.
2) To obtain an accurate estimation of the low-rank component, we introduce the ε-shrinkage TNN (ε-TNN) regularization into the proposed model. The ε-TNN constraint on the coefficient tensor adaptively penalizes the tensor singular values with different weights, which effectively addresses the overshrinkage problem.
3) We devise an effective iterative optimization algorithm based on the alternating direction method of multipliers (ADMM) to solve the proposed TLRA-MSL model. Extensive experimental results prove the effectiveness and superiority of the proposed method in HAD task.

The framework of this article is arranged as follows. First, we present the necessary definitions of the tensor notations and preliminaries used in our method in Section II. In Section III, we introduce the proposed TLRA-MSL model in detail.

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Algorithm 1 Tensor Product

Input: $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, $\mathcal{B} \in \mathbb{R}^{n_2 \times n_4 \times n_3}$.

1: $\overline{\mathcal{A}} = \text{fft}(\mathcal{A}, [\ ], 3)$ and $\overline{\mathcal{B}} = \text{fft}(\mathcal{B}, [\ ], 3)$.

2: Calculate each frontal slice of $\bar{\mathcal{C}}$ by

$$\bar{\mathcal{C}}^{(i)} = \begin{cases} \bar{\mathcal{A}}^{(i)} \cdot \bar{\mathcal{B}}^{(i)}, & i = 1, \ldots, \lceil \frac{n_3+1}{2} \rceil, \\ \text{conj}\left(\bar{\mathcal{C}}^{(n_3-i+2)}\right), & i = \lceil \frac{n_3+1}{2} \rceil + 1, \ldots, n_3. \end{cases}$$

3: $\mathcal{C} = \text{ifft}(\bar{C}, [\ ], 3)$.

Output: $\mathcal{C} = \mathcal{A} * \mathcal{B} \in \mathbb{R}^{n_1 \times n_4 \times n_3}$.

In Section IV, the experimental results and analysis are provided. Finally, the conclusions are summarized in Section V.

II. Notations and Preliminaries

In this section, we present all the notations and preliminaries used in this article. We remark the real numbers and complex numbers as $\mathbb{R}$ and $\mathbb{C}$, respectively. Meanwhile, we denote an m-dimensional tensor as $\mathcal{A} \in \mathbb{C}^{n_1 \times n_2 \times \cdots n_m}$. The notations of tensor are denoted in the Nomenclature. Note that the matrix and tensor in the Nomenclature denote special cases of the m-dimensional tensor when $m = 2$ and $m = 3$. Next, we introduce some definitions about tensor operation for developing the TLRA-MSL model. First, for a 3-D HSI tensor $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, the unfold and corresponding inverse operation are denoted as follows:

$$\text{unfold}(\mathcal{A}) = \begin{bmatrix} \mathcal{A}^{(1)} \\ \mathcal{A}^{(2)} \\ \vdots \\ \mathcal{A}^{(n_3)} \end{bmatrix}, \quad \text{fold}\ (\text{unfold}(\mathcal{A})) = \mathcal{A} \tag{1}$$

where the unfold operation indicates the tensor matricization operation, which maps $\mathcal{A}$ to a matrix with a size of $n_1 n_3 \times n_2$. In addition, the block circulant matrix $\text{bcirc}(\mathcal{A}) \in \mathbb{R}^{n_1 n_3 \times n_2 n_3}$ is anchored as follows:

$$\text{bcirc}(\mathcal{A}) = \begin{bmatrix} \mathcal{A}^{(1)} & \mathcal{A}^{(n_3)} & \cdots & \mathcal{A}^{(2)} \\ \mathcal{A}^{(2)} & \mathcal{A}^{(1)} & \cdots & \mathcal{A}^{(3)} \\ \vdots & \vdots & \ddots & \vdots \\ \mathcal{A}^{(n_3)} & \mathcal{A}^{(n_3-1)} & \cdots & \mathcal{A}^{(1)} \end{bmatrix}. \tag{2}$$

The following lines present some basic definitions of tensor algebra used in this article.

Definition 1 (Tensor Conjugate Transpose [57]): The conjugate transpose tensor $\mathcal{A}^* \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ corresponding to the original tensor $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ is computed by reversing the sequence of frontal slices about $\mathcal{A}$ from 2 to n_3 and transposing each reversed frontal slice, i.e.,

$$\begin{aligned} \left(\mathcal{A}^*\right)^{(1)} &= \left(\mathcal{A}^{(1)}\right)^T \\ \left(\mathcal{A}^*\right)^{(i)} &= \left(\mathcal{A}^{(n_3+2-i)}\right)^T, \quad i = 2, 3, \ldots, n_3. \end{aligned} \tag{3}$$

Definition 2 (Tensor Product [58]): Let $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ and $\mathcal{B} \in \mathbb{R}^{n_2 \times n_4 \times n_3}$. Then, the tensor product $\mathcal{C} = \mathcal{A} * \mathcal{B} \in \mathbb{R}^{n_1 \times n_4 \times n_3}$ is defined as follows:

$$\mathcal{A} * \mathcal{B} = \text{fold}(\text{bcirc}(\mathcal{A}) \cdot \text{unfold}(\mathcal{B})). \tag{4}$$

The tensor product can also be computed in an efficient way, as shown in Algorithm 1, which is based on the fast Fourier transform (FFT) operation in MATLAB. If each frontal slice of a tensor is diagonal, it can be called f-diagonal tensor. Besides, a tensor $\mathcal{Q} \in \mathbb{R}^{n_1 \times n_1 \times n_3}$ is an orthogonal tensor if it satisfies $\mathcal{Q} * \mathcal{Q}^* = \mathcal{Q}^* * \mathcal{Q} = \mathcal{I}$.

Definition 3 (t-SVD and Skinny t-SVD [59]): If $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, the t-SVD of tensor $\mathcal{A}$ is expressed as follows:

$$\mathcal{A} = \mathcal{U} * \mathcal{S} * \mathcal{V}^* \tag{5}$$

where $\mathcal{S} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ is f-diagonal and $\mathcal{U} \in \mathbb{R}^{n_1 \times n_1 \times n_3}$ and $\mathcal{V} \in \mathbb{R}^{n_2 \times n_2 \times n_3}$ are orthogonal tensors. The Skinny t-SVD of tensor $\mathcal{A}$ is denoted as $\mathcal{A} = \mathcal{U}_s * \mathcal{S}_s * \mathcal{V}_s^*$, where $\mathcal{U}_s = \mathcal{U}(:, 1 : \text{rank}_t(\mathcal{A}), :)$, $\mathcal{S}_s = \mathcal{S}(1 : \text{rank}_t(\mathcal{A}), 1 : \text{rank}_t(\mathcal{A}), :)$, and $\mathcal{V}_s = \mathcal{V}(:, 1 : \text{rank}_t(\mathcal{A}), :)$ where $\text{rank}_t(\mathcal{A})$ denotes the tensor tubal rank of $\mathcal{A}$ (see (6) in Definition 4).

Definition 4 (Tensor Average Rank and Tubal Rank [60]): For a tensor $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ and corresponding t-SVD, i.e., $\mathcal{A} = \mathcal{U} * \mathcal{S} * \mathcal{V}^*$, the average rank of tensor $\mathcal{A}$ is denoted as $\text{rank}_a(\mathcal{A}) = (\sum_{i=1}^{n_3} r_i)/n_3$, where $(r_1, \ldots, r_{n_3}) = (\text{rank}(\bar{\mathcal{A}}^{(1)}), \ldots, \text{rank}(\bar{\mathcal{A}}^{(n_3)}))$. The tensor tubal rank $\text{rank}_t(\mathcal{A})$ of $\mathcal{A}$ denotes the number of nonzero singular tubes of $\mathcal{S}$, which can be anchored as follows:

$$\text{rank}_t(\mathcal{A}) = \sum_{k=1}^{\min(n_1, n_2)} \mathbb{I}_{\mathcal{S}(k,k,:) \neq 0} \tag{6}$$

where $\mathcal{S}(k, k, :)$ is the kth tube of $\mathcal{S}$. $\mathbb{I}_{\mathcal{S}(k,k,:)}$ denotes the indicator function, where $\mathbb{I}_{\mathcal{S}(k,k,:)} = 1$ if $\mathcal{S}(k, k, :) \neq 0$ and $\mathbb{I}_{\mathcal{S}(k,k,:)} = 0$ if $\mathcal{S}(k, k, :) = 0$.

Definition 5 (TNN [61]): Let $\|\mathcal{A}\|_*$ be the TNN of $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, and then, it can be defined as follows:

$$\|\mathcal{A}\|_* = \frac{1}{n_3} \sum_{k=1}^{n_3} \left\|\bar{\mathcal{A}}^{(k)}\right\|_*. \tag{7}$$

Definition 6 (ε-Shrinkage Thresholding Operator [62]): Given a scalar $x \in \mathbb{R}$, the ε-shrinkage thresholding operator is denoted as

$$g_\varepsilon^\theta(x) = \text{sgn}(x) \max\left\{|x| - \theta |x|^{\varepsilon - 1}, 0\right\} \tag{8}$$

where scalars $\theta \in \mathbb{R}$ and $\varepsilon \in \mathbb{R}$ satisfy $\theta > 0$ and $\varepsilon \leq 1$, respectively.

III. Proposed Method

The detection flowchart of the proposed TLRA-MSL is presented in Fig. 1. First, for a 3-D hyperspectral data cube, we perform the orthogonal transformation such as PCA and maximum noise fraction (MNF) transformation to reduce the spectral dimension and computational load. Then, the preprocessed HSI tensor is factorized by Skinny t-SVD to construct the background dictionary. Finally, we develop the TLRA-MSL model and corresponding ADMM solver to isolate the sparse anomaly component from heterogeneous backgrounds.

A. HAD Via the TLRA-MSL Model

Let the preprocessed HSI tensor be denoted as $\mathcal{H} \in \mathbb{R}^{w \times h \times b}$, where w and h denote the width and height of each frontal slice, respectively, and b denotes the amounts of reduced spectral bands. Therefore, the HSI tensor $\mathcal{H}$ can be decomposed

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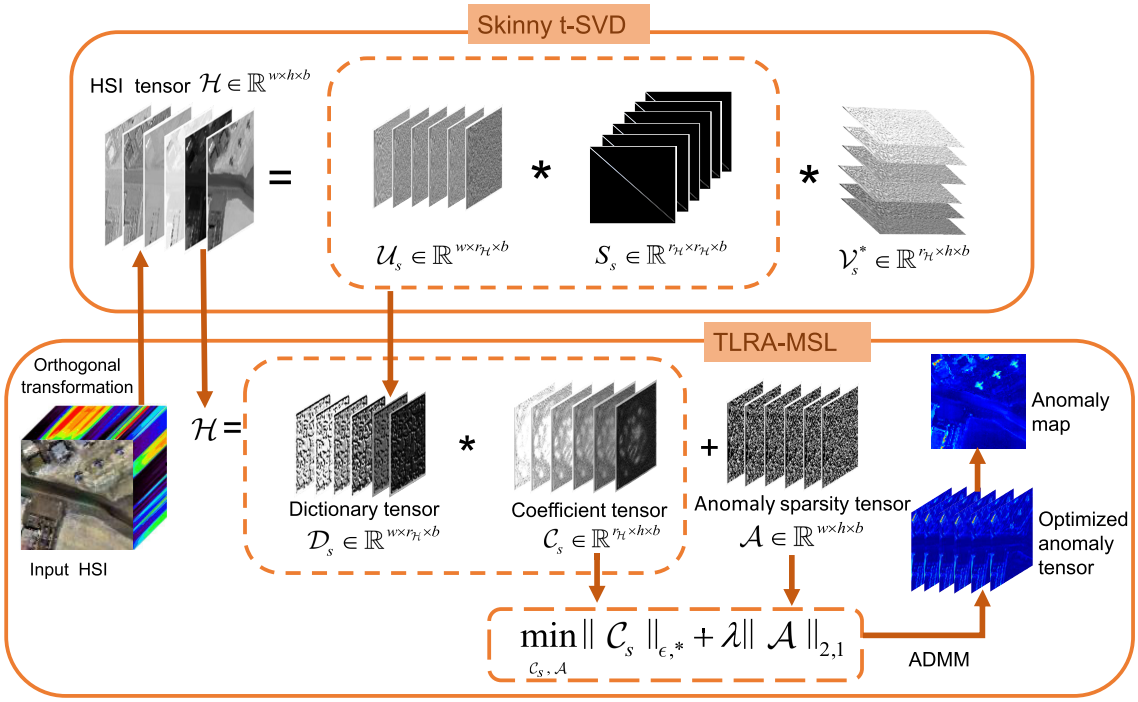


Fig. 1. Schematic of the proposed TLRA-MSL model.

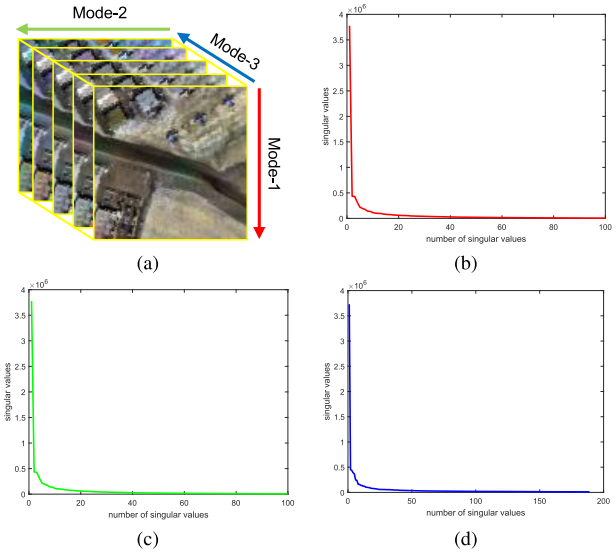


Fig. 2. Diagrammatic sketch of low-rank attribute in the background tensor. (a) Background tensor. (b) Singular value statistics of mode-1 unfolding matrix. (c) Singular value statistics of mode-2 unfolding matrix. (d) Singular value statistics of mode-3 unfolding matrix.

into a background component and an anomalous component, which is anchored as follows:

$$\mathcal{H} = \mathcal{B} + \mathcal{A} \tag{9}$$

where $\mathcal{A}, \mathcal{B} \in \mathbb{R}^{w\times h\times b}$ represent the anomaly tensor and background tensor, respectively. The observation model in (9) well preserves the inherent 3-D tensor structure without any tensor matricization operation. Meanwhile, it can be observed from Fig. 2 that the singular values of the mode-i ($i = 1, 2, 3$) unfolding matrices drop to zero rapidly, which demonstrates the low-rank characteristic of the background tensor. However, existing HAD methods mostly presume that the HSI background is derived from a single low-rank subspace. In practice, however, the heterogeneous background of an HSI may come from multiple subspaces because of the variable shooting angle of the hyperspectral imager. In fact, the multisubspace attribute of HSI was verified in [46], where it was postulated that the background tensor consists of multiple low-rank subspaces, and a vector linear representation was utilized in the matrix domain to represent the low-rank component in HSIs. In our method, we promote the multisubspace property of HSIs to a higher dimensional tensor domain and employ the tensor linear representation strategy for multiple subspace learning (MSL) and better 3-D tensor data characterization. Specifically, the background tensor $\mathcal{B}$ in (9) can be represented by dictionary tensor $\mathcal{D}$ and corresponding coefficient tensor $\mathcal{C}$, which is denoted as the following TLRA model:

$$\min_{\mathcal{C}} \operatorname{rank}_t(\mathcal{C}), \qquad \text{s.t. } \mathcal{B} = \mathcal{D} * \mathcal{C}. \tag{10}$$

To obtain the low-rank representation of the background, the dictionary tensor $\mathcal{D}$ should include the bases of all background subspaces. Inspired by the self-representation model [63], [64], [65], we can select the original HSI tesnor $\mathcal{H}$ itself as a dictionary, i.e., $\mathcal{B} = \mathcal{H} * \mathcal{C}_{\mathcal{H}}$, where $\mathcal{C}_{\mathcal{H}} \in \mathbb{R}^{h\times h\times b}$ represents the corresponding coefficient tensor. Nevertheless, for higher algorithm efficiency and lower computational burden, we opt to upgrade the construction of dictionary tensor by calculating the Skinny t-SVD of the HSI tensor $\mathcal{H}$, i.e., $\mathcal{H} = \mathcal{U}_s * \mathcal{S}_s * \mathcal{V}_s^*$. As a result, the background tensor $\mathcal{B}$ in (10) can be transformed to $\mathcal{B} = \mathcal{U}_s * \mathcal{S}_s * \mathcal{V}_s^* * \mathcal{C}_{\mathcal{H}}$. Then, we denote $\mathcal{U}_s * \mathcal{S}_s$ and $\mathcal{V}_s^* * \mathcal{C}_{\mathcal{H}}$ as dictionary tensor $\mathcal{D}_s \in \mathbb{R}^{w\times r_{\mathcal{H}}\times b}$ and coefficient tensor $\mathcal{C}_s \in \mathbb{R}^{r_{\mathcal{H}}\times h\times b}$, where the tensor tubal rank of $\mathcal{H}$ satisfies $r_{\mathcal{H}} \ll h$. Therefore, in each iteration, the computation of SVD for matrices will decrease from $h \times h$ to $r_{\mathcal{H}} \times h$. Finally,

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the coefficient tensor $\mathcal{C}_s$ becomes the latent variable required to be optimized, which is initialized to zeros [see [65] for a detailed proof of the equivalence of minimizer pair $(\mathcal{C}_s, \mathcal{A})$ and minimizer pair $(\mathcal{C}_\mathcal{H}, \mathcal{A})$]. Furthermore, considering that the anomalies always appear in the analogous spatial location in the spectral band direction, we can employ group sparsity regularization to characterize the spatial–spectral sparsity of anomalies. The tensor group sparsity $l_{2,1}$ norm is utilized to constrain the anomalous tensor $\mathcal{A}$, which is defined as follows:

$$\begin{aligned}\|\mathcal{A}\|_{2,1} &= \sum_{i=1}^{w}\sum_{j=1}^{h}\|\mathcal{A}(i,j,:)\|_2 \\ &= \sum_{i=1}^{w}\sum_{j=1}^{h}\sqrt{\sum_{k=1}^{b}\mathcal{A}(i,j,k)^2}\end{aligned} \tag{11}$$

where $\|\cdot\|_2$ denotes the l_2 norm of vector (see the Nomenclature).

B. Accurate Low-Rank Estimation With ε-TNN

For the problem in (10), the tensor tubal rank function $\text{rank}_t(\cdot)$ is adopted for the low-rank estimation of the coefficient tensor $\mathcal{C}_s$, which makes the optimization problem of (10) nonconvex and NP-hard [66]. Therefore, many researchers have exploited the TNN to approximate the tensor tubal rank, which transforms the low-rank estimation issue in (10) into a convex optimization problem. Nevertheless, as a convex envelope of the tensor tubal rank, TNN still exists a fatal flaw of overshrinkage [67], which means that all singular values will be imposed by the same penalties. In other words, the importance of singular values is not well reflected. To tackle this problem, we develop a tighter surrogate of the tensor tubal rank named the ε-TNN, which adaptively penalizes the singular values with the ε-shrinkage thresholding operator described in Definition 6. As shown in Fig. 3, we present several representative ε-shrinkage thresholding functions $g_\varepsilon^\theta(x)$ with the same threshold $\theta = 1$ but different ε values. It can be seen that if $\varepsilon = 1$, ε-shrinkage thresholding operator will degenerate into the soft thresholding, which imposes the same penalties for all inputs and is utilized in the optimization of the TNN. In contrast, if $\varepsilon \to$-inf, there is no penalty for inputs larger than θ, which is transformed into a hard thresholding operation. When ε satisfies -inf $< \varepsilon < 1$, the vertical coordinate distance between the ε-shrinkage thresholding function (dotted line) and the hard thresholding function (black solid line) will gradually reduce as the input increases. Therefore, if -inf $< \varepsilon < 1$, the ε-shrinkage thresholding can impose fewer penalties for larger inputs of singular values, which effectively preserves the high-quality information of a large singular value.

This property is beneficial to obtain a more accurate low-rank estimation of the coefficient tensor. After incorporating the ε-shrinkage thresholding operator to the TNN, the ε-TNN for the coefficient tensor $\mathcal{C}_s$ can be denoted as follows:

$$\|\mathcal{C}_s\|_{\varepsilon,*} = \frac{1}{b}\sum_{k=1}^{b}\left\|\overline{\mathcal{C}_s}^{(k)}\right\|_{\varepsilon,*} = \frac{1}{b}\sum_{k=1}^{b}\sum_{i=1}^{\min(r_\mathcal{H},h)} G_\varepsilon^\theta(\overline{\mathcal{S}}) \tag{12}$$

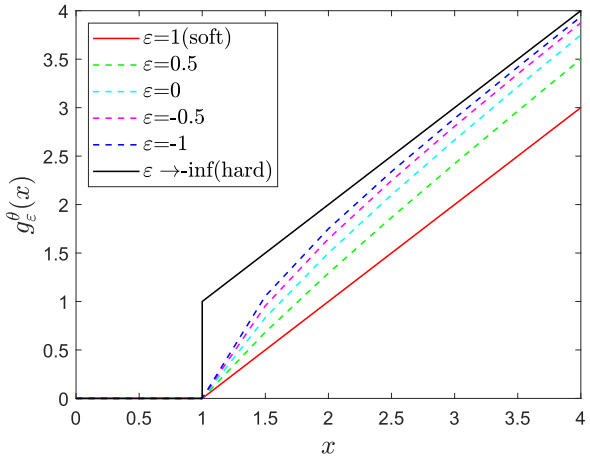


Fig. 3. ε-shrinkage functions when ε is set to 1, 0.5, 0, −0.5, −1, and −infinity. θ is set to 1. The larger ε value represents a greater penalty on the input.

Algorithm 2 ε-Shrinkage Tensor Singular Value Thresholding

Input: $\mathcal{C}_s, \mathcal{Q} \in \mathbb{R}^{r_\mathcal{D}\times h\times b}$, θ, ε

$\overline{\mathcal{Q}} = \text{fft}(\mathcal{Q}, [\,], 3)$;
for $i = 0, 1, \ldots, \left[\frac{b+1}{2}\right]$ **do**
  $[\overline{\mathcal{U}}(:,:,i), \overline{\mathcal{S}}(:,:,i), \overline{\mathcal{V}}(:,:,i)] = \text{SVD}(\overline{\mathcal{Q}}(:,:,i))$;
  Compute $\overline{\mathcal{G}}(:,:,i)$ by Eq. (13)
    $\overline{\mathcal{G}}(:,:,i) = \max\left\{\left|\overline{\mathcal{S}}(:,:,i)\right| - \theta\left|\overline{\mathcal{S}}(:,:,i)\right|^{\varepsilon-1}, 0\right\}$;
  $\overline{\mathcal{W}}(:,:,i) = [\overline{\mathcal{U}}(:,:,i), \overline{\mathcal{G}}(:,:,i), \overline{\mathcal{V}}(:,:,i)]$;
end for
for $i = \left[\frac{b+1}{2}+1\right], \ldots, b$ **do**
  $\overline{\mathcal{W}}(:,:,i) = \text{conj}(\overline{\mathcal{W}}(:,:,b-i+2))$;
end for

Output: $\xi(\mathcal{Q}) = \text{ifft}(\overline{\mathcal{W}}, [\,], 3)$

where $\overline{\mathcal{S}}$ is the front slice of f-diagonal tensor $\mathcal{S}$ and $\mathcal{S}$ is obtained from the t-SVD of $\mathcal{C}_s$, i.e., $\mathcal{C}_s = \mathcal{U} * \mathcal{S} * \mathcal{V}^*$. Here, $G_\varepsilon^\theta(\overline{\mathcal{S}})$ can be formulated as follows:

$$\begin{aligned}\left(G_\varepsilon^\theta(\overline{\mathcal{S}})\right)_{iik} &= g_\varepsilon^\theta\left(\overline{\mathcal{S}}_{iik}\right) \\ &= \max\left\{\left|\overline{\mathcal{S}}_{iik}\right| - \theta\left|\overline{\mathcal{S}}_{iik}\right|^{\varepsilon-1}, 0\right\}\end{aligned} \tag{13}$$

where $\overline{\mathcal{S}}_{iik}$ denotes the singular value of each front slice $\overline{\mathcal{C}_s}(:,:,k)$. To optimize the problem of ε-TNN, we define the following proximal operator:

$$\text{Prox}_{\gamma\|\mathcal{C}_s\|_{\varepsilon,*}}(\mathcal{Q}) = \arg\min_{\mathcal{C}_s}\gamma\|\mathcal{C}_s\|_{\varepsilon,*} + \frac{1}{2}\|\mathcal{C}_s - \mathcal{Q}\|_F^2. \tag{14}$$

According to [68], if $\mathcal{U} * \mathcal{G} * \mathcal{V}^*$ be the t-SVD of $\mathcal{Q}$, the proximal operator in (14) can be addressed by the following solution: $\text{Prox}_{\gamma\|\mathcal{C}_s\|_{\varepsilon,*}}(\mathcal{Q}) = \mathcal{U}*\mathcal{G}*\mathcal{V}^*$, where γ and f-diagonal tensor $\mathcal{G} \in \mathbb{R}^{r_\mathcal{H}\times h\times b}$ meet the conditions of $\gamma > 0$ and $\overline{\mathcal{G}} = G_\varepsilon^\theta(\overline{\mathcal{S}})$, respectively. The solution of the optimization problem in (14) is described as an ε-shrinkage tensor singular value thresholding operation $\xi(\mathcal{Q})$, which is presented in Algorithm 2.

C. Optimization Procedure

By incorporating the ε-TNN and group sparsity $l_{2,1}$ norm into our method, the TLRA-MSL model can be reformulated

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as follows:

$$\begin{aligned}&\min_{\mathcal{C}_s,\mathcal{A}} \|\mathcal{C}_s\|_{\varepsilon,*}+\lambda\|\mathcal{A}\|_{2,1}\\&\text{s.t. } \mathcal{H}=\mathcal{D}_s*\mathcal{C}_s+\mathcal{A}\end{aligned}\tag{15}$$

where λ represents the regularization parameter. To optimize the above problem, we first introduce an auxiliary variable $\mathcal{R}$ to separate the object $\|\mathcal{C}_s\|_{\varepsilon,*}$ from constraint $\mathcal{H}=\mathcal{D}_s*\mathcal{C}_s+\mathcal{A}$. Therefore, the optimization problem (15) can be described as follows:

$$\begin{aligned}&\min_{\mathcal{C}_s,\mathcal{A}} \|\mathcal{C}_s\|_{\varepsilon,*}+\lambda\|\mathcal{A}\|_{2,1}\\&\text{s.t. } \mathcal{C}_s=\mathcal{R},\ \mathcal{H}=\mathcal{D}_s*\mathcal{R}+\mathcal{A}.\end{aligned}\tag{16}$$

The above optimization problem can be effectively solved by the ADMM [69] algorithm. Then, we introduce the inexact augmented Lagrangian multiplier (IALM) to tackle the problem in (16), which is denoted as follows:

$$\begin{aligned}\mathcal{L}(\mathcal{C}_s,\mathcal{R},\mathcal{A},\mathcal{M}_1,\mathcal{M}_2)=&\ \|\mathcal{C}_s\|_{\varepsilon,*}+\lambda\|\mathcal{A}\|_{2,1}\\&+\langle\mathcal{M}_1,\mathcal{C}_s-\mathcal{R}\rangle+\frac{\mu}{2}\|\mathcal{C}_s-\mathcal{R}\|_F^2\\&+\langle\mathcal{M}_2,\mathcal{H}-\mathcal{D}_s*\mathcal{R}-\mathcal{A}\rangle\\&+\frac{\mu}{2}\|\mathcal{H}-\mathcal{D}_s*\mathcal{R}-\mathcal{A}\|_F^2\end{aligned}\tag{17}$$

where μ denotes the penalty parameter and $\mathcal{M}_1$ and $\mathcal{M}_2$ denote the Lagrangian multiplier. The optimization problem in (17) can be divided into a range of subproblems and solved by ADMM, which updates one variable with others fixed. The specific optimization steps are presented as follows.

1) Update $\mathcal{C}_s$: The objective function with regard to $\mathcal{C}_s$ can be denoted as

$$\begin{aligned}\mathcal{C}_s^{k+1}&=\operatorname*{argmin}_{\mathcal{C}_s}\|\mathcal{C}_s\|_{\varepsilon,*}+\left\langle\mathcal{M}_1^k,\mathcal{C}_s-\mathcal{R}^k\right\rangle+\frac{\mu^k}{2}\left\|\mathcal{C}_s-\mathcal{R}^k\right\|_F^2\\&=\operatorname*{argmin}_{\mathcal{C}_s}\|\mathcal{C}_s\|_{\varepsilon,*}+\frac{\mu^k}{2}\left\|\mathcal{C}_s-\left(\mathcal{R}^k-\mathcal{M}_1^k/\mu^k\right)\right\|_F^2\end{aligned}\tag{18}$$

which is equivalent to the optimization subproblem in (14) and can be solved by the ε-shrinkage tensor singular value thresholding solver. Let $\mathcal{Q}^k=\mathcal{R}^k-\mathcal{M}_1^k/\mu^k$ and $\gamma^k=1/\mu^k$, and (18) can be solved by $\mathcal{C}_s^{k+1}=\xi(\mathcal{Q}^k)$ according to Algorithm 2.

2) Update $\mathcal{A}$: The objective function with regard to $\mathcal{A}$ can be denoted as

$$\begin{aligned}\mathcal{A}^{k+1}&=\operatorname*{argmin}_{\mathcal{A}}\lambda\|\mathcal{A}\|_{2,1}+\left\langle\mathcal{M}_2^k,\mathcal{H}-\mathcal{D}_s*\mathcal{R}^k-\mathcal{A}\right\rangle\\&\quad+\frac{\mu^k}{2}\left\|\mathcal{H}-\mathcal{D}_s*\mathcal{R}^k-\mathcal{A}\right\|_F^2\\&=\operatorname*{argmin}_{\mathcal{A}}\lambda\|\mathcal{A}\|_{2,1}\\&\quad+\frac{\mu^k}{2}\left\|\mathcal{A}-\left(\mathcal{H}-\mathcal{D}_s*\mathcal{R}^k+\mathcal{M}_2^k/\mu^k\right)\right\|_F^2.\end{aligned}\tag{19}$$

Let $\mathcal{T}^k=\mathcal{H}-\mathcal{D}_s*\mathcal{R}^k+\mathcal{M}_2^k/\mu^k$, and the optimization subproblem in (19) can be solved by the $l_{2,1}$ minimization operator according to [70], which is described as follows:

$$\mathcal{A}^{k+1}(i,j,:)=\max\left(1-\frac{\lambda}{\mu^k\|\mathcal{T}^k(i,j,:)\|_2},0\right)\cdot\mathcal{T}^k(i,j,:).\tag{20}$$

Algorithm 3 TLRA-MSL for HAD

Input: Input preprocessed HSI data $\mathcal{H}\in\mathbb{R}^{w\times h\times b}$, parameters ε, μ, λ

1: **Initialization:** $\mathcal{C}_s^0=\mathcal{R}^0=\mathcal{M}_1^0=0$, $\mathcal{A}^0=\mathcal{M}_2^0=0$, $\mu_0=1e-4$, $\mu_{\max}=1e8$, iteration number $k=1$, max iteration number $\varphi=100$; $\eta=1.1$, $\phi=10^{-8}$, $\mathcal{D}_s=\mathcal{U}_s*\mathcal{S}_s$ with Skinny t-SVD: $\mathcal{H}=\mathcal{U}_s*\mathcal{S}_s*\mathcal{V}_s^*$,
2: **while** $0\le k\le\varphi$ or not converged **do**
3: update $\mathcal{C}_s$ by solving Eq. (18) via ε-shrinkage tensor singular value thresholding operator;
4: update $\mathcal{A}$ by Eq. (19);
5: update $\mathcal{R}$ by Eq. (21);
6: update the multipliers $\mathcal{M}_1$, $\mathcal{M}_2$ by Eq. (23);
7: update the penalty parameter $\mu=\min(\gamma\mu,\mu_{\max})$;
8: update $k=k+1$;
9: check the convergence conditions

$$\max\left\{\begin{array}{c}\left\|\mathcal{C}_s^k-\mathcal{C}_s^{k-1}\right\|_\infty\\\left\|\mathcal{A}^k-\mathcal{A}^{k-1}\right\|_\infty\\\left\|\mathcal{R}^k-\mathcal{R}^{k-1}\right\|_\infty\\\left\|\mathcal{C}_s^k-\mathcal{R}^k\right\|_\infty\\\left\|\mathcal{H}-\mathcal{D}_s*\mathcal{R}^k-\mathcal{A}^k\right\|_\infty\end{array}\right\}\le\phi$$

10: **end while**
11: compute the anomaly map $\mathbf{O}(i,j)=\|\mathcal{A}(i,j,:)\|_2$;
Output: anomaly map $\mathbf{O}$

3) Update $\mathcal{R}$: The objective function with regard to $\mathcal{R}$ can be denoted as

$$\begin{aligned}\mathcal{R}^{k+1}&=\operatorname*{argmin}_{\mathcal{R}}\left\langle\mathcal{M}_1^k,\mathcal{C}_s^{k+1}-\mathcal{R}\right\rangle+\frac{\mu^k}{2}\left\|\mathcal{C}_s^{k+1}-\mathcal{R}\right\|_F^2\\&\quad+\left\langle\mathcal{M}_2^k,\mathcal{H}-\mathcal{D}_s*\mathcal{R}-\mathcal{A}^{k+1}\right\rangle\\&\quad+\frac{\mu^k}{2}\left\|\mathcal{H}-\mathcal{D}_s*\mathcal{R}-\mathcal{A}^{k+1}\right\|_F^2\\&=\operatorname*{argmin}_{\mathcal{R}}\left\|\mathcal{C}_s^{k+1}+\mathcal{M}_1^k/\mu^k-\mathcal{R}\right\|_F^2\\&\quad+\left\|\mathcal{H}-\mathcal{A}^{k+1}+\mathcal{M}_2^k/\mu^k-\mathcal{D}_s*\mathcal{R}\right\|_F^2.\end{aligned}\tag{21}$$

This problem can be solved by taking the derivative of (21) at $\mathcal{R}$ as zero to obtain the solution. Therefore, the optimization solution of (21) can be described as follows:

$$\mathcal{R}^{k+1}=\left(\mathcal{D}_s^**\mathcal{D}_s+\mathcal{I}\right)^{-1}*\left(\mathcal{D}_s^**\mathcal{N}_1^k+\mathcal{N}_2^k\right)\tag{22}$$

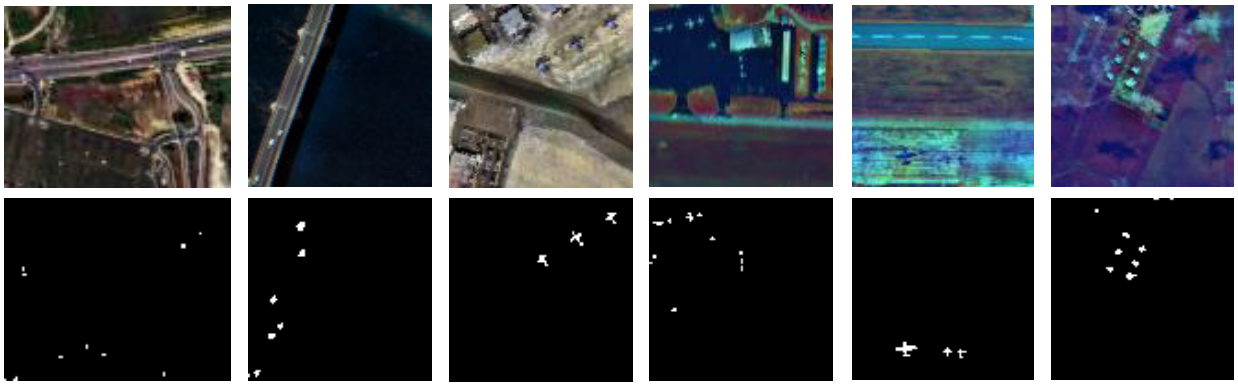
where $\mathcal{N}_1^k=\mathcal{H}-\mathcal{A}^{k+1}+\mathcal{M}_2^k/\mu^k$, $\mathcal{N}_2^k=\mathcal{C}_s^{k+1}+\mathcal{M}_1^k/\mu^k$, and $(\cdot)^{-1}$ denotes the tensor inverse, which satisfies $(\cdot)*(\cdot)^{-1}=\mathcal{I}$.

4) Update $\mathcal{M}_1$, $\mathcal{M}_2$, and μ: The penalty parameter μ and the multipliers $\mathcal{M}_1$ and $\mathcal{M}_2$ are updated as follows:

$$\begin{cases}\mathcal{M}_1^{k+1}=\mathcal{M}_1^k+\mu^k(\mathcal{C}_s^{k+1}-\mathcal{R}^{k+1})\\\mathcal{M}_2^{k+1}=\mathcal{M}_2^k+\mu^k(\mathcal{H}-\mathcal{D}_s*\mathcal{R}^{k+1}-\mathcal{A}^{k+1})\\\mu^{k+1}=\min\left(\eta\mu^k,\mu_{\max}\right).\end{cases}\tag{23}$$

After multiple iterations to reach convergence conditions, the anomaly map $\mathbf{O}$ can be obtained. The detailed optimization procedure of TLRA-MSL can be found in Algorithm 3. The computational complexity and convergence analysis will be discussed in Section IV-F.

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(a) (b) (c) (d) (e) (f)

Fig. 4. Six datasets used in our experiments. The first row is the pseudo-color image of each dataset and the second row is the corresponding ground truth. (a) HYDICE. (b) Pavia. (c) San Diego. (d) Gainesville. (e) Gulfport. (f) Texas Coast.

IV. Experimental Results and Analyses

In this section, a sequence of experiments are conducted to validate the anomaly detection performance of the proposed method on six well-known hyperspectral imagery datasets. All the experiments are conducted using a personal computer with MATLAB R2019a, Windows 10, Intel[1] Core[2] i7-9750H CPU@2.60 GHz, and 16-GB RAM.

A. Datasets Description

In the experiments, we verify the detection performance of the proposed TLRA-MSL on six HSI datasets. The detailed descriptions of these datasets are described as follows.

1) HYDICE Dataset [32]: This dataset is an urban scene captured by the airborne sensor named Hyperspectral Digital Imagery Collection Experiment (HYDICE). The spatial resolution and spectral resolution of this dataset can reach 1 m and 10 nm, respectively. As shown in Fig. 4(a), an image subarea with a size of 80 × 100 pixels and 175 spectral bands is cropped from the original image with a size of 370 × 370 pixels for the experiments. There are 21 anomalous pixels, which are presented as cars and roofs in complex backgrounds.

2) Pavia Dataset [40]: This dataset was from the Pavia city center in northern Italy and captured by an airborne sensor named the Reflective Optics System Imaging Spectrometer (ROSIS). The spatial resolution of this dataset reaches 1.3 m/pixel. Meanwhile, this dataset is composed of an area of 80 × 100 pixels with 102 spectral bands in the wavelength range of 430–860 nm. There exist 43 vehicles and bare soil anomalous pixels embedded in the bridge deck and water surface, which account for 0.54% of all images. The pseudo-color image and ground truth of this dataset are shown in Fig. 4(b).

3) San Diego Dataset [71]: This dataset is an airport scene captured by an airborne sensor named the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS). It was shot from San Diego with a 3.5-m/pixel spatial resolution. The wavelengths of the spectral bands range from 370 to 2510 nm. The anomalous pixels and background pixels in the San Diego dataset are mainly composed of airplanes and airport zones. The cropped spatial size of this dataset is 100 × 100 pixels, and there are 189 bands. As shown in Fig. 4(c), anomalous aircraft targets are represented by 134 anomalous pixels to be detected in the San Diego dataset.

4) Gainesville Dataset [72]: This dataset is an urban scene, which is also captured by the AVIRIS airborne sensor over Gainesville, FL, USA. By removing the water absorption and low signal-to-noise ratio (SNR) bands from the original data, 191 bands were preserved for experiments. Meanwhile, the spatial resolution and size of this dataset are 3.5 m/pixels and 100 × 100 pixels, respectively. As shown in Fig. 4(d), this dataset contains 156 anomaly pixels, which account for 1.56% of all images, to be explored.

5) Gulfport Dataset [72]: This dataset was shot by the AVIRIS airborne hyperspectral sensor over the Gulfport, Southern Mississippi, USA, with a 3.4-m/pixel spatial resolution and 191 spectral bands in the wavelength range of 400–2500 nm. The spatial size of this dataset can reach up to 100 × 100 pixels. As shown in Fig. 4(e), there are 180 anomalous airplane pixels, which account for 1.8% of all image pixels. The background mainly consists of vegetation, highways, and airport runways.

6) Texas Coast Dataset [72]: This urban dataset was shot by the AVIRIS airborne hyperspectral sensor over the Texas Coast, TX, USA, in 2010. The spatial resolution and size of this scene are 17.2 m/pixel and 100 × 100 pixels, respectively. There are 201 anomalous pixels and 204 spectral bands in the wavelength range of 450–1350 nm. The pseudo-color image and ground truth of this dataset are shown in Fig. 4(f).

B. Evaluation Indicators and Comparison Methods

1) Evaluation Indicators: In the experiments, we employ the detection accuracy and detection speed to evaluate the performance of each HAD algorithm. The run times of each algorithm on six datasets can be seen as the detection speed evaluation indicator. For detection accuracy, we exploit several qualitative and quantitative evaluation indicators to evaluate the detection performance. Specifically, we present the detection map, 2-D receiver operating characteristic (ROC) curve, 3-D ROC curve, and anomaly background separation map as the qualitative evaluation indicators in the experiments. The detection maps can intuitively show the abnormal detection effect. The 2-D ROC curve shows the curve of

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TABLE I
PARAMETER SETTINGS OF THE TLRA-MSL MODEL

Data Sets	ε	θ	b_{PCA}	b_{MNF}	λ
HYDICE	-0.5	1	8	26	
Pavia	-1	1	5	5	
San Diego	-1	9	6	14	0.01
Gainesville	-1	9	16	43	
Gulfport	-1	3	17	23	
Texas Coast	-1	3	9	27	

the detection probability P_D versus the false alarm (FA) probability P_F with the change in the separation threshold $\tau \in (0, 1)$. Here, P_D and P_F can be calculated as follows:

$$P_D = \frac{N_D}{N_T}, \quad P_F = \frac{N_F}{N_A} \tag{24}$$

where N_D and N_T represent the quantities of detected true anomalous pixels and all true anomalous pixels under a certain τ, respectively, and N_F and N_A represent the quantities of detected false anomalous pixels and all pixels in the image under a threshold τ, respectively. In addition, the 3-D ROC curves incorporate the threshold τ to construct a 3-D evaluation curve that reflects the comprehensive detectability and the background suppression capacity of each HAD algorithm.

The anomaly background separation map can intuitively display the separation degree of the background and anomaly components by illustrating the box plot of each component. The larger the gap between the background box and anomaly box is, the better the separation and detection performance. In addition to qualitative evaluation indicators, we introduce several quantitative evaluation indicators according to the 3-D ROC curves. By fixing one of the three variables P_D, P_D, and τ, we can obtain a 2-D ROC curve of (P_D, P_F), a 2-D ROC curve of (P_D, τ), and a 2-D ROC curve of (P_F, τ). Then, three quantitative evaluation indicators, i.e., $\text{AUC}_{(P_D,P_F)}$, $\text{AUC}_{(P_D,\tau)}$, and $\text{AUC}_{(P_F,\tau)}$, can be derived by calculating the area under the curve (AUC) of the corresponding ROC curves. The higher the $\text{AUC}_{(P_D,P_F)}$ and $\text{AUC}_{(P_D,\tau)}$ values are, the better the detection performance. The smaller the $\text{AUC}_{(P_F,\tau)}$ value is, the worse the background suppression performance. Based on these evaluation indicators, we can further employ the comprehensive indicators AUC_{OD} and AUC_{SNPR}, which are defined as follows:

$$\text{AUC}_{\text{OD}} = \text{AUC}_{(P_D,P_F)} + \text{AUC}_{(P_D,\tau)} - \text{AUC}_{(P_F,\tau)} \tag{25}$$

$$\text{AUC}_{\text{SNPR}} = \frac{\text{AUC}_{(P_D,\tau)}}{\text{AUC}_{(P_F,\tau)}} \tag{26}$$

where AUC_{OD} can be seen as an overall evaluation indicator to measure the detection performance and AUC_{SNPR} represents the comprehensive signal-to-noise probability ratio (SNPR) indicator. The higher the AUC_{OD} and AUC_{SNPR} values are, the better the detection performance of the HAD algorithm.

2) Comparison Methods: To better compare the proposed TLRA-MSL with other state-of-the-art HAD methods, we choose eight representative HAD algorithms as comparison methods, including GRXD [18], CRD [32], FrFTRX [73], LRASR [71], TPCA [50], GTVLRR [43], PTA [56], and PCA-TLRAR [55]. The GRXD [18] is a benchmark statistical theory-based model. The FrFTRX [73] is considered as a variant and promotion GRXD [18] model. CRD [32] is a classic model based on collaborative representation. The LRASR [71], GTVLRR [43], and PTA [56] are three state-of-the-art HAD models based on low-rank and sparse matrix factorization. TPCA [50] and PCA-TLRAR [55] are representative tensor-based low-rank and sparse factorization methods. The parameters of the eight comparison methods are tuned to their optimum settings according to the corresponding literature. The parameter settings of the proposed TLRA-MSL model on the six HSI datasets are presented in Table I. b_{PCA} and b_{MNF} represent the numbers of bands after PCA and MNF orthogonal transformations, respectively. The regularization parameter λ in the TLRA-MSL model is fixed to 0.001 in our experiments. ε and θ represent the parameters in the ε-shrinkage thresholding operator. As shown in Table I, b_{PCA} and b_{MNF} are empirically tuned to the best settings for each dataset. We also conducted some parameter tuning experiments for ε, θ, and λ. The detailed parameter tuning results of these three parameters will be discussed in Section IV-C.

C. Detection Performance Comparison

1) Detection Speed Comparison: We first compare the execution time of the proposed method with other state-of-the-art HAD methods. The running times of different models on the six HSI datasets are reported in Table II. We observe that the GRXD speeds the minimum running time on the six HSI datasets. In contrast, the GTVLRR takes the longest execution time on all the datasets. The execution time of the proposed TLRA-MSL model is only longer than that of GRXD, which demonstrates the efficiency of our method.

2) Detection Accuracy Comparison: Then, we carry out a series of detection accuracy comparison experiments. The qualitative experimental results are presented in Figs. 5–8. First, the detection maps of the nine HAD algorithms on the six datasets are shown in Fig. 5. Note that the color in Fig. 5 changes from dark blue to dark red, which indicates that the anomalous degree of the corresponding region is increasing. The detection maps of the nine HAD methods on the HYDICE dataset are presented in the first row of Fig. 5. Anomalies can be highlighted based on the responses of GTVLRR, FrFTRX, LRASR, and TLRA-MSL. However, fewer FAs are generated using our method and GTVLRR than the other methods. Meanwhile, our algorithm runs faster than GTVLRR. The anomaly maps of these algorithms on the Pavia dataset are presented in the second row of Fig. 5. The PTA, LRASR, and CRD achieve lower response to anomaly targets compared to GTVLRR, PCA-TLRAR, and the proposed TLRA-MSL. Some background contamination is observed when using GRXD, TPCA, and FrFTRX. The anomaly maps of these algorithms on the San Diego dataset are presented in the third row of Fig. 5. The GRXD, CRD, LRASR, and TPCA are unsuccessful in detecting the abnormal airplane. The detection results of PTA and PCA-TLRAR are seriously contaminated by the backgrounds. The GTVLRR and our method can well detect the anomalies, and the proposed TLRA-MSL has a

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TABLE II
RUNNING TIME OF NINE HAD ALGORITHMS ON THE SIX HSI DATASETS

Data Sets	TLRA-MSL	GRXD [18]	CRD [32]	FrFTRX [73]	LRASR [71]	TPCA [50]	GTVLRR [43]	PTA [56]	PCA-TLRAR [55]
HYDICE	1.6943	0.0495	27.4459	10.7568	18.76201	11.5195	119.6324	13.7937	8.0576
Pavia	0.2585	0.0237	21.1011	9.6140	18.5925	8.2624	66.9908	11.3243	9.4487
San Diego	1.5506	0.0658	41.6217	16.3128	21.6446	15.5859	102.9572	19.5589	9.3339
Gainesville	5.7613	0.0625	41.1696	16.2340	26.1348	17.2291	118.5813	20.4448	9.2939
Gulfport	6.2734	0.0598	42.1911	16.0997	23.5081	17.4394	111.9387	19.7621	9.3669
Texas Coast	3.1636	0.0619	46.4023	16.7985	25.5766	17.5201	145.4880	21.6990	9.3852

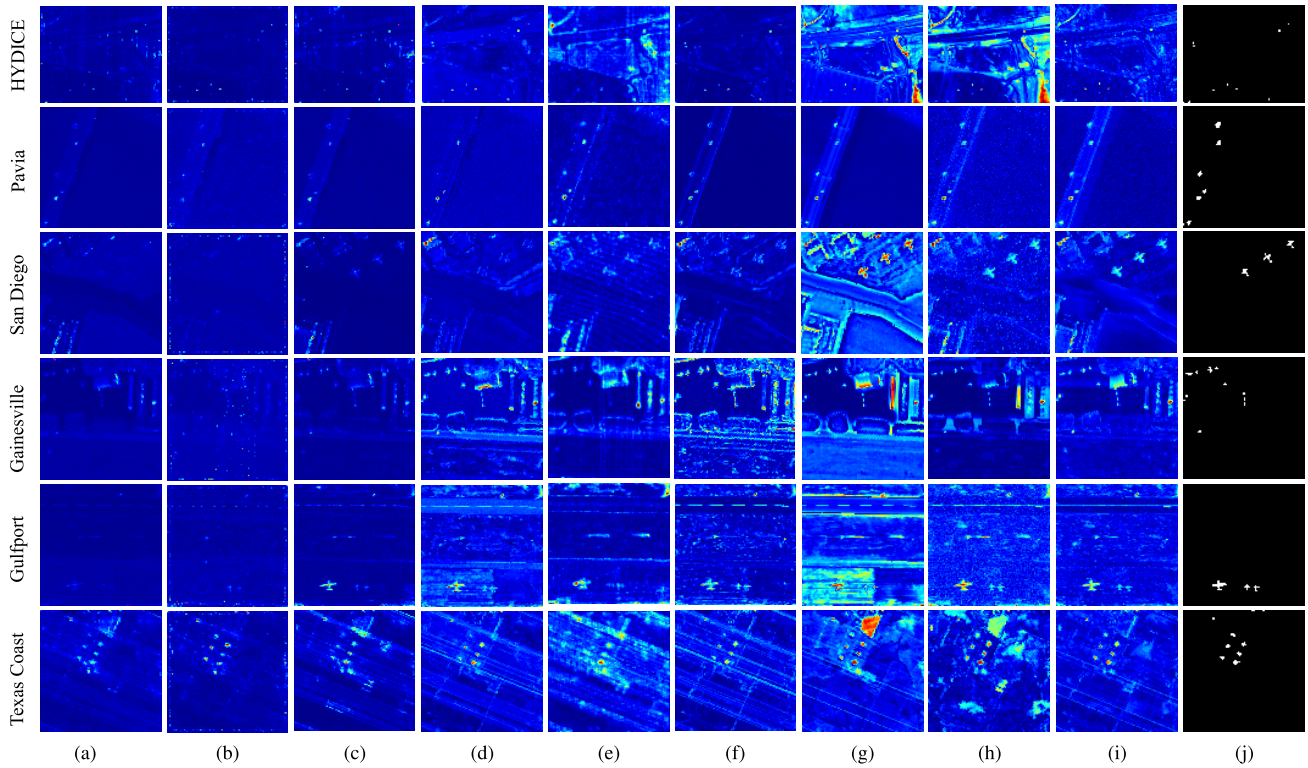


Fig. 5. Color anomaly maps derived by different HAD algorithms and the corresponding ground truth on six datasets. (a) GRXD. (b) CRD. (c) FrFTRX. (d) LRASR. (e) TPCA. (f) GTVLRR. (g) PTA. (h) PCA-TLRAR. (i) TLRA-MSL. (j) Ground truth.

stronger background suppression effect compared to GTVLRR on this dataset. The detection maps of these nine algorithms on the Gainesville dataset are shown in the fourth row of Fig. 5. The PTA and TPCA are unsuccessful in detecting the abnormal target in the urban scene. The GRXD, CRD, and FrFTRX have low responses to small abnormal targets. The LRASR, GTVLRR, PCA-TLRAR, and our method can well detect the small abnormal pixels compared to other methods. The detection maps of these HAD methods on the Gulfport dataset are shown in the fifth row of Fig. 5. We can observe that the CRD and PTA fail to detect three abnormal airplanes in this scene. The GRXD can only detect one of the larger airplanes. The detection result of TPCA misses the basic shape information of the abnormal airplanes. The LRASR and PTA engender a lot of FAs, which inevitably contaminates the response maps. The GTVLRR, FrFTRX, and PCA-TLRAR achieve better detection maps compared to other methods. The anomaly maps of these algorithms on the Texas Coast dataset are shown in the sixth row of Fig. 5. The TPCA is unsuccessful in detecting the abnormal targets. The other methods well separate the abnormal components from the background, but CRD, PTA, PCA-TLRAR, and FrFTRX generate more FAs pixels. Meanwhile, the GTVLRR, LRASR, GRXD, and TLRA-MSL have higher discrimination between background and anomalies in the anomaly maps. In addition, as shown in Figs. 6 and 7, we present the 2-D ROC curve of (P_D, P_F) and 3-D ROC curve of the nine HAD algorithms on the six HSI datasets. It can be seen that the proposed TLRA-MSL model yields a higher P_D for the same P_F in most situations, which demonstrates the superiority of our method in the HAD task compared to other methods. Furthermore, as shown in Fig. 8, we present the normalized anomaly background separation maps to reflect the degree of discrimination between the anomaly component and background component in the detection results. The blue box and red box in Fig. 8 represent the data distributions of the background and anomaly components in the detection maps, respectively. Specifically, the larger the gap between the two boxes, the better the separation performance of the detector. It can be seen from Fig. 8(a) that LRASR offers the best separation between the background and anomalies, and the separation performance of our method follows closely. On the

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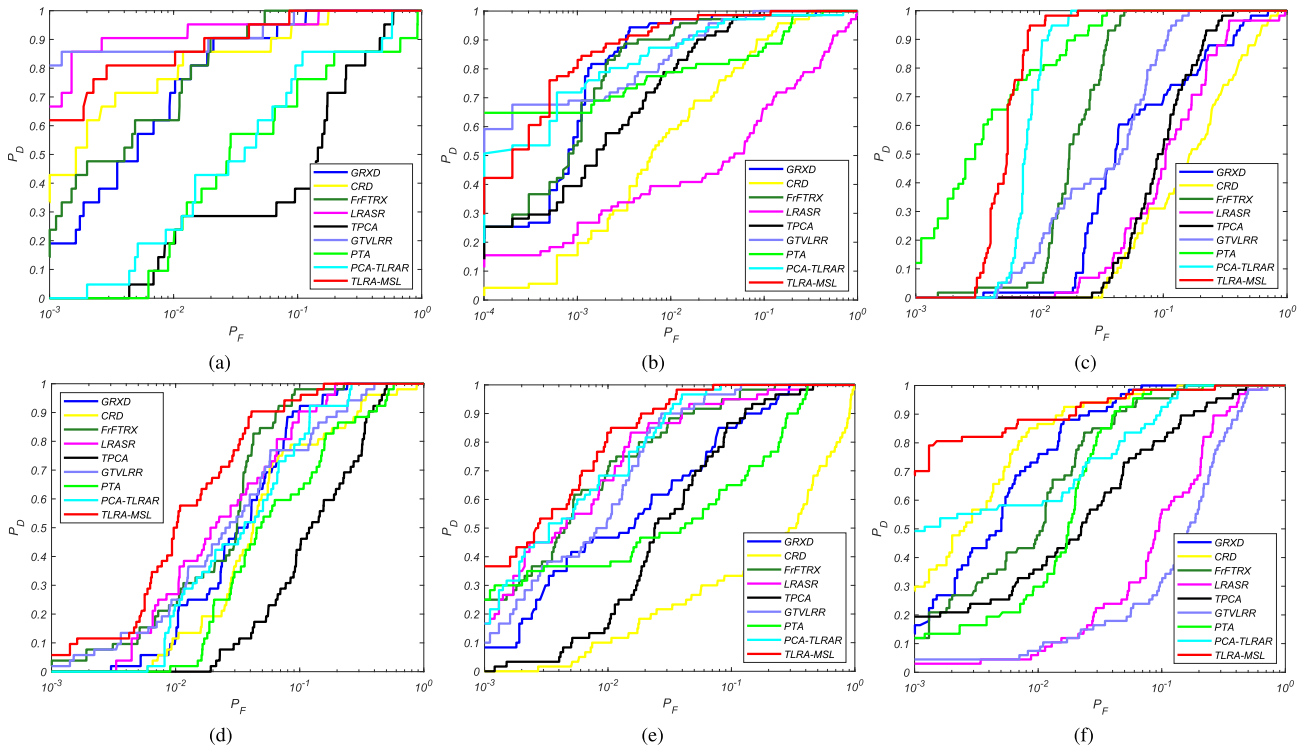


Fig. 6. 2-D (P_D, P_F) ROC curves of nine HAD algorithms on the six HSI datasets. (a) HYDICE. (b) Pavia. (c) San Diego. (d) Gainesville. (e) Gulfport. (f) Texas Coast.

Pavia, San Diego, and Gainesville datasets, the proposed TLRA-MSL obtain the largest gaps between background and anomalous box, whereas other methods occur smaller gaps or more overlap compared to our method. However, we can observe in Fig. 8(f) that all methods have poor performance on the Texas Coast dataset. In addition, as shown in Fig. 8(e), GTVLRR, PCA-TLRAR, and our method achieve similar optimal separation performance on this dataset. Overall, the proposed TLRA-MSL can reach the best performance in terms of the separation between the anomalies and background on most of the datasets.

In addition to the above qualitative results, we also present the quantitative experimental results of different methods to directly measure the detection performance. As shown in Table III, we record the five different metrics of AUC in view of the 3-D ROC curve, and the optimal and suboptimal results are denoted by bold and underline number, respectively. It can be perceived from Table III that the proposed TLRA-MSL obtains the best performance of $\mathrm{AUC}_{(P_D,P_F)}$ on all datasets. Meanwhile, our method also gets the highest $\mathrm{AUC}_{(P_D,\tau)}$ on the HYDICE and Pavia datasets. For the comprehensive detection performance metric $\mathrm{AUC}_{\mathrm{ON}}$, our method achieve the optimal results on the HYDICE, Pavia, and Gainesville datasets. Then, the suboptimal results of $\mathrm{AUC}_{\mathrm{ON}}$ on the San Diego and Gulfport datasets are also obtained by our method. In addition, our method also reaches the optimal results of $\mathrm{AUC}_{\mathrm{SNPN}}$ on the Gainesville dataset and the suboptimal results of $\mathrm{AUC}_{\mathrm{SNPN}}$ on the Pavia and San Diego dataset. These quantitative experimental results effectively prove the superiority of our method in the HAD task. In view of these qualitative and quantitative experimental results, we summarize that our TLRA-MSL obtains the most comprehensive and credible detection performance over the comparison models.

D. Parameter Tuning and Analysis

As shown in Table I, we record the optimal parameter settings of the proposed TLRA-MSL model. First, for the setting of bands after PCA and MNF, i.e., b_{PCA} and b_{MNF}, we perform a search from 1 to 50 and choose the optimal value as the one that yields the best performance. The optimal b_{PCA} and b_{MNF} values are reported in Table I. It can be seen that b_{PCA} is always lower than b_{MNF} in every dataset. By comparing the detection performance between these two orthogonal transformation methods, we find that PCA is more suitable for the proposed TLRA-MSL because of its better detection speed and accuracy. Therefore, we choose b_{PCA} as the number of bands for the follow-up parameter tuning experiments. Given space limitations, we will not list the parameter tuning results of b_{PCA} and b_{MNF}. We focus on the parameter settings of ε, θ, and regularization parameter λ.

1) Impact of ε and θ: ε and θ are two key parameters in the ε-TNN, which determines the effect of accurate low-rank estimation. These two parameters satisfy -inf $< \varepsilon < 1$ and $\theta > 0$. In our experiments, to explore the impact of ε and θ on the HAD performance, we set $\varepsilon \in \{1, 3, 5, 7, 9\}$ and $\theta \in \{-1, -0.5, 0, 0.5, 1\}$ on the six datasets. As shown in Fig. 9, we present the different $\mathrm{AUC}_{(P_D,P_F)}$ values with different combinations of ε and θ on the six datasets. It can be observed that the highest $\mathrm{AUC}_{(P_D,P_F)}$ values are obtained when $\theta = 9$ and $\varepsilon = -1$ on the San Diego and Gainesville datasets. Meanwhile, the optimal combination of ε and θ is set to $(-1, 3)$ on the Gulfport and Texas Coast datasets.

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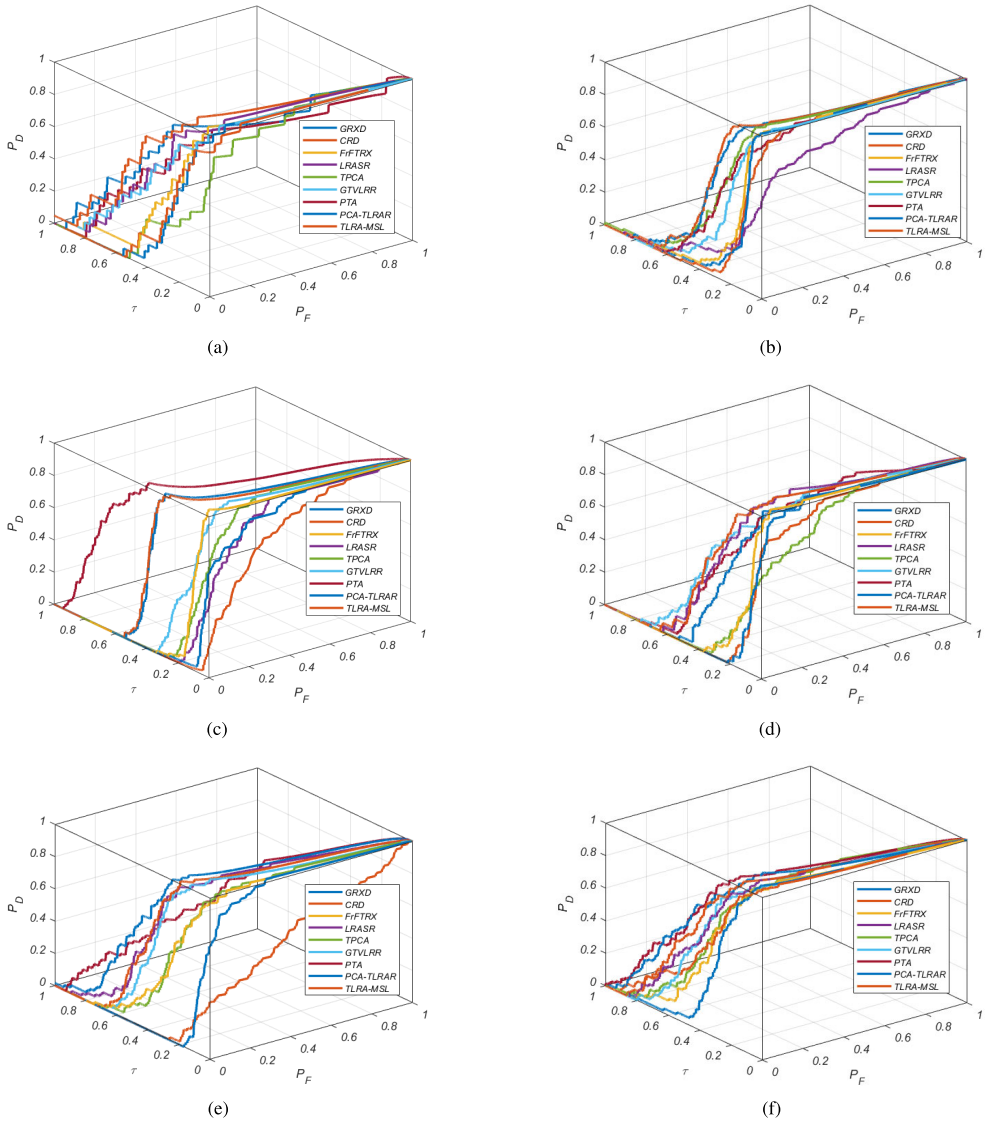


Fig. 7. 3-D ROC curves of nine HAD methods on the six HSI datasets. (a) HYDICE. (b) Pavia. (c) San Diego. (d) Gainesville. (e) Gulfport. (f) Texas Coast.

For the HYDICE dataset, the optimal combination of ε and θ is $(-0.5, 1)$. The detailed settings of θ and ε are reported in Table I.

2) Impact of λ: The regularization parameter λ is a momentous parameter, which controls the balance between low-rank and sparse components during decomposition in (15). As shown in Fig. 10, we choose the parameter λ from the set $\{0.0001, 0.0005, 0.001, 0.005, 0.01, 0.05, 0.1\}$. It can be seen that the $\text{AUC}_{(P_D,P_F)}$ values gradually increase and obtain the peak value at 0.01 when λ varies from 0.0001 to 0.01 on the HYDICE and Gainesville datasets. For the Pavia and San Diego datasets, $\text{AUC}_{(P_D,P_F)}$ values are relatively stable and slightly fluctuate with λ. Meanwhile, the $\text{AUC}_{(P_D,P_F)}$ values achieved on the Gulfport and Texas coast datasets also obtain the optimal performance when $\lambda = 0.01$. Based on the above considerations, we set λ as 0.01 for the proposed TLRA-MSL.

E. Ablation Study

To investigate the contributions of three pivotal components, i.e., orthogonal transformation, Skinny t-SVD, and ε-TNN, on the proposed method, we conduct an ablation study on the six datasets. The specific experimental results are recorded in Tables IV and V.

1) With and Without Orthogonal Transformation: To analyze the effect of orthogonal transformation on the proposed TLRA-MSL, we record the performance of our method with and without orthogonal transformation in Tables IV and V, where the amount of increase and decrease compared with the complete TLRA-MSL model is marked in red and blue, respectively. In addition, OT and Skinny in Tables IV and V represent the orthogonal transformation and Skinny t-SVD, respectively. It can be observed from Table IV that the $\text{AUC}_{(P_D,P_F)}$ values obtained by the proposed TLRA-MSL without orthogonal transformation decrease to varying degrees on all the datasets. As presented in Table V, the running times of the proposed TLRA-MSL without orthogonal transformation greatly increase on all datasets. Based on these experimental results, we can conclude that orthogonal transformation plays a prominent role in the accuracy and efficiency improvements of our method.

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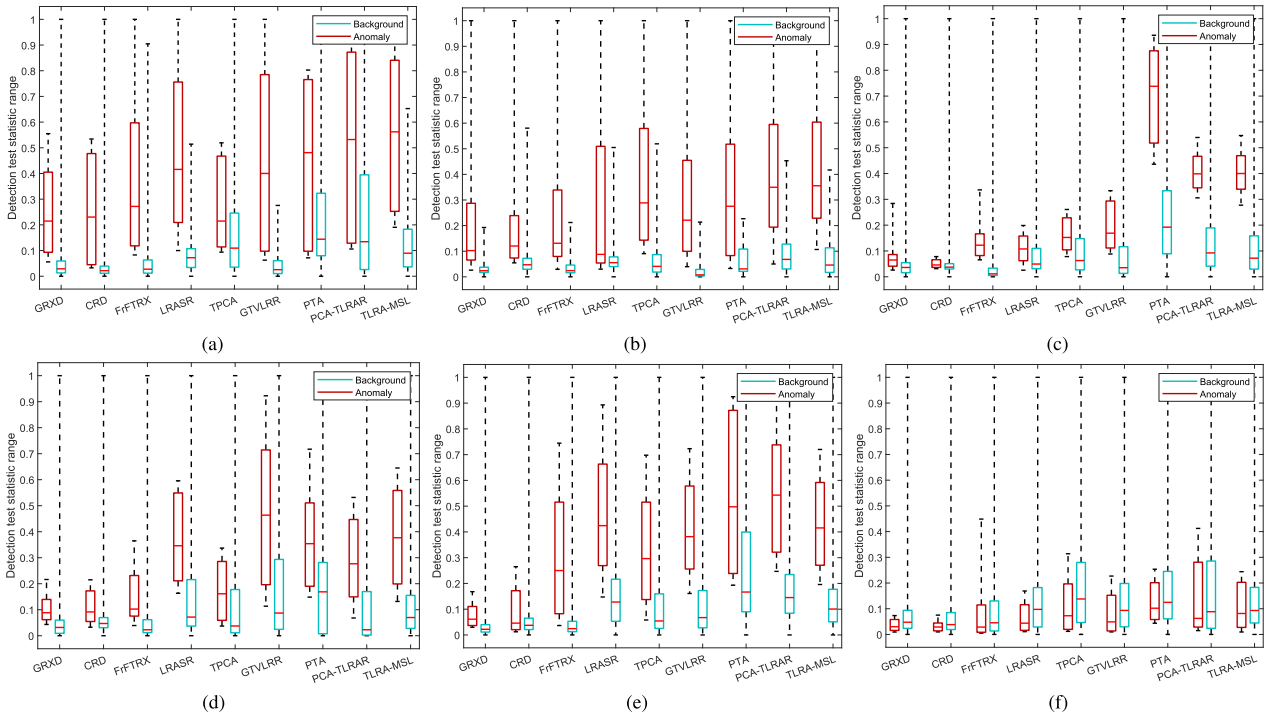


Fig. 8. Anomaly background separation maps of the nine HAD methods on the six HSI datasets. (a) HYDICE. (b) Pavia. (c) San Diego. (d) Gainesville. (e) Gulfport. (f) Texas Coast.

TABLE III
AUC VALUES OF NINE HAD ALGORITHMS ON THE SIX HSI DATASETS

Data Sets	AUC Values	GRXD [18]	CRD [32]	FrFTRX [73]	LRASR [71]	TPCA [50]	GTVLRR [43]	PTA [56]	PCA-TLRAR [55]	TLRA-MSL
HYDICE	$\text{AUC}(P_D, P_F)$	0.9856	0.9822	0.9902	0.9919	0.8241	0.9920	0.8377	0.8899	**0.9922**
	$\text{AUC}(P_D, \tau)$	0.2404	0.2624	0.3322	0.4762	0.2655	0.4625	0.4734	0.5556	**0.5691**
	$\text{AUC}(P_F, \tau)$	0.0351	**0.0259**	0.0359	0.0727	0.1295	0.0321	0.1796	0.1803	0.1028
	AUC_{OD}	1.1909	1.2187	1.2865	1.3954	0.9601	1.4224	1.1315	1.2652	**1.4365**
	AUC_{SNPR}	6.8440	10.1042	9.2491	6.5458	2.0494	**14.3949**	2.6348	3.0815	5.5062
Pavia	$\text{AUC}(P_D, P_F)$	0.9934	0.9664	0.9935	0.8338	0.9922	0.9948	0.9697	0.9858	**0.9973**
	$\text{AUC}(P_D, \tau)$	0.1600	0.1554	0.1895	0.1872	0.3322	0.2636	0.3048	0.3746	**0.3854**
	$\text{AUC}(P_F, \tau)$	0.0257	0.0500	0.0276	0.0575	0.0489	**0.0138**	0.0451	0.0751	0.0567
	AUC_{OD}	1.1155	1.0593	1.1438	0.9495	1.2659	1.2340	1.2190	1.2762	**1.3171**
	AUC_{SNPR}	5.7445	2.8675	6.4454	3.0525	6.5974	**18.3366**	6.5252	4.8664	6.6420
San Diego	$\text{AUC}(P_D, P_F)$	0.8885	0.7638	0.9787	0.8410	0.8848	0.9511	0.9933	0.9915	**0.9948**
	$\text{AUC}(P_D, \tau)$	0.0685	0.0487	0.1270	0.1085	0.1584	0.1939	**0.7230**	0.4019	0.4255
	$\text{AUC}(P_F, \tau)$	0.0381	0.0423	**0.0193**	0.0631	0.0805	0.0540	0.2078	0.1085	0.0904
	AUC_{OD}	0.9191	0.7703	1.0866	0.8867	0.9629	1.0912	**1.5087**	1.2852	1.3300
	AUC_{SNPR}	1.8033	1.1530	**6.5971**	1.7222	1.9699	3.5951	3.4799	3.7074	4.7092
Gainesville	$\text{AUC}(P_D, P_F)$	0.9512	0.9005	0.9663	0.9565	0.8224	0.9337	0.8827	0.9398	**0.9854**
	$\text{AUC}(P_D, \tau)$	0.0962	0.1004	0.1363	0.3565	0.1604	**0.4621**	0.3547	0.2790	0.4529
	$\text{AUC}(P_F, \tau)$	0.0350	0.0521	**0.0327**	0.1060	0.0728	0.1289	0.1565	0.0596	0.0940
	AUC_{OD}	1.0126	0.9491	1.0701	1.2071	0.9101	1.2672	1.0810	1.1593	**1.3444**
	AUC_{SNPR}	2.7527	1.9320	4.1751	3.3634	2.2037	3.5878	2.2672	4.6861	**4.8192**
Gulfport	$\text{AUC}(P_D, P_F)$	0.9525	0.6336	0.9821	0.9781	0.9432	0.9847	0.8957	0.9889	**0.9927**
	$\text{AUC}(P_D, \tau)$	0.0736	0.0698	0.2868	0.4507	0.2981	0.4027	0.5319	**0.5403**	0.4391
	$\text{AUC}(P_F, \tau)$	**0.0247**	0.0433	0.0324	0.1358	0.0789	0.0887	0.2100	0.1569	0.1112
	AUC_{OD}	1.0014	0.6600	1.2365	1.2930	1.1624	1.2986	1.2177	**1.3723**	1.3207
	AUC_{SNPR}	2.9742	1.6093	**8.8311**	3.3175	3.7768	4.5376	2.5330	3.4422	3.9498
Texas Coast	$\text{AUC}(P_D, P_F)$	0.9906	0.9919	0.9814	0.9858	0.9390	0.9777	0.9775	0.9749	**0.9923**
	$\text{AUC}(P_D, \tau)$	0.3142	0.4461	0.3943	0.4847	0.4412	0.4557	**0.6018**	0.5878	0.5294
	$\text{AUC}(P_F, \tau)$	0.0555	**0.0480**	0.0629	0.1028	0.1521	0.1046	0.1498	0.1218	0.1011
	AUC_{OD}	1.2388	1.3815	1.3033	1.3597	1.2187	1.3202	1.4231	**1.4343**	1.4133
	AUC_{SNPR}	5.4699	**9.1104**	6.1155	4.6383	2.8446	4.2768	3.9763	4.7741	5.1634

2) With and Without Skinny t-SVD: The ablation study of Skinny t-SVD is also presented in Tables IV and V. In our method, we introduce the Skinny t-SVD instead of t-SVD to obtain the background dictionary $\mathcal{D}_s$ more efficiently. To verify this viewpoint, we conduct an ablation study of Skinny t-SVD in Tables IV and V. It is observed that the $\text{AUC}_{(P_D, P_F)}$

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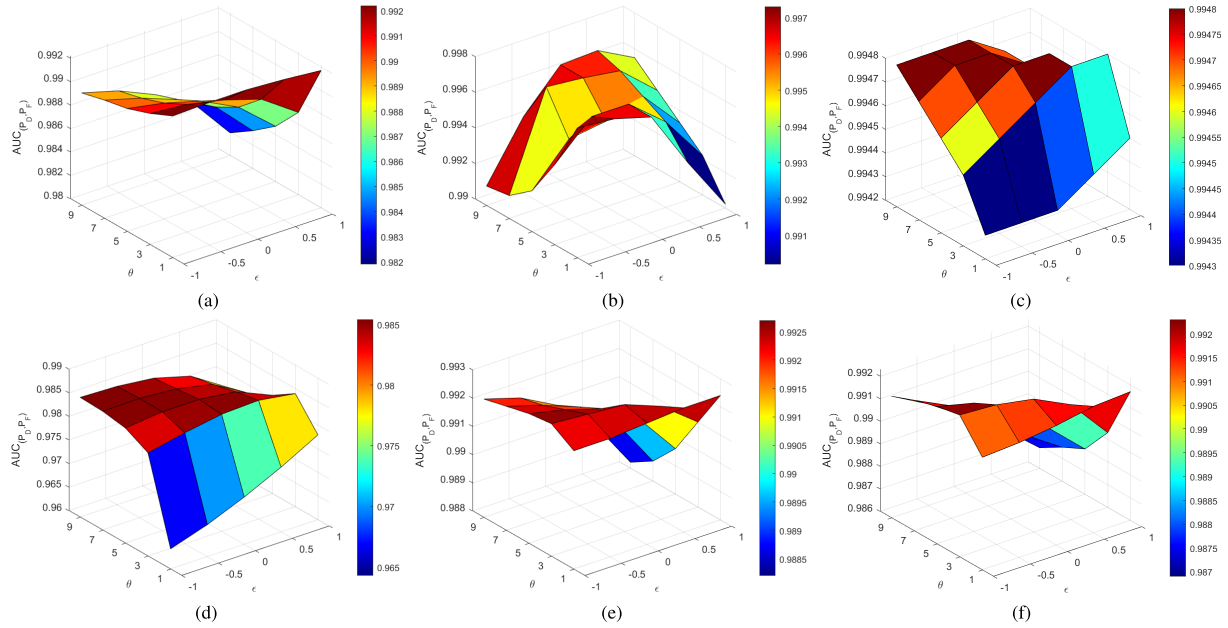


Fig. 9. $\mathrm{AUC}_{(P_D,P_F)}$ of the proposed TLRA-MSL on the six datasets with different combinations of ε and θ. (a) HYDICE. (b) Pavia. (c) San Diego. (d) Gainesville. (e) Gulfport. (f) Texas Coast.

TABLE IV

ABLATION STUDY OF THE PROPOSED TLRA-MSL ON THE SIX DATASETS OVER $\mathrm{AUC}_{(P_D,P_F)}$ VALUES (%)

Models	HYDICE	Pavia	San Diego	Gainesville	Gulfport	Texas Coast
TLRA-MSL without OT	97.27 (-1.95)	92.73 (-7.00)	95.31 (-4.17)	95.17 (-3.37)	99.13 (-0.14)	96.51 (-2.72)
TLRA-MSL without Skinny	99.48 (+0.26)	99.17 (-0.56)	99.50 (+0.02)	99.00 (+0.46)	98.25 (-1.02)	97.93 (-1.30)
TLRA-MSL without ε-TNN	99.00 (-0.22)	92.72 (-7.01)	99.47 (-0.01)	98.43 (-0.11)	99.18 (-0.09)	99.10 (-0.13)
TLRA-MSL	99.22	99.73	99.48	98.54	99.27	99.23

TABLE V

ABLATION STUDY OF THE PROPOSED TLRA-MSL ON THE SIX DATASETS OVER RUNNING TIME (S)

Models	HYDICE	Pavia	San Diego	Gainesville	Gulfport	Texas Coast
TLRA-MSL without OT	46.2510 (+44.5567)	44.8551 (+44.5966)	68.4920 (+66.9414)	82.4718 (+76.7105)	81.4759 (+75.2025)	71.3290 (+68.1651)
TLRA-MSL without Skinny	1.8045 (+0.1102)	1.8821 (+1.6236)	1.8816 (+0.3310)	6.0067 (+0.2454)	6.5004 (+0.2270)	3.2888 (+0.1252)
TLRA-MSL without ε-TNN	1.6010 (-0.0933)	0.3143 (+0.0558)	1.4659 (-0.0847)	5.4228 (-0.3385)	5.8528 (-0.4206)	2.8904 (-0.2732)
TLRA-MSL	1.6943	0.2585	1.5506	5.7613	6.2734	3.1636

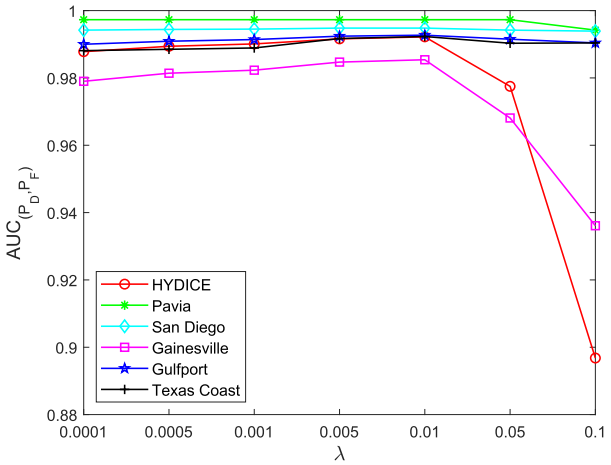


Fig. 10. $\mathrm{AUC}_{(P_D,P_F)}$ change curve with respect to λ on the six datasets.

values obtained by the proposed TLRA-MSL with t-SVD fluctuate slightly around the $\mathrm{AUC}_{(P_D,P_F)}$ values obtained by the proposed TLRA-MSL with Skinny t-SVD. However, the running times derived from the proposed TLRA-MSL with Skinny t-SVD are all lower than the running times derived from the proposed TLRA-MSL with t-SVD, which proves that the employment of Skinny t-SVD can effectively improve the efficiency of our method.

3) With and Without ε-TNN: In Section III-B, we apply the ε-TNN to estimate the accurate low-rank coefficient tensor. To investigate the contribution of ε-TNN on our method, we conduct ablation experiments on ε-TNN. The TLRA-MSL without ε-TNN represents the TLRA-MSL model with the TNN. We compare these two models over $\mathrm{AUC}_{(P_D,P_F)}$ values and running times. From the experimental results, we can observe that the $\mathrm{AUC}_{(P_D,P_F)}$ values obtained by the TLRA-MSL model with TNN are lower than those of the TLRA-MSL model with ε-TNN on all datasets. The running times obtained by the TLRA-MSL model with TNN fluctuate slightly around the running times obtained by the TLRA-MSL model based on ε-TNN. Therefore, we can conclude from this ablation study that the performance of the proposed method can be effectively boosted by the implementation of ε-TNN.

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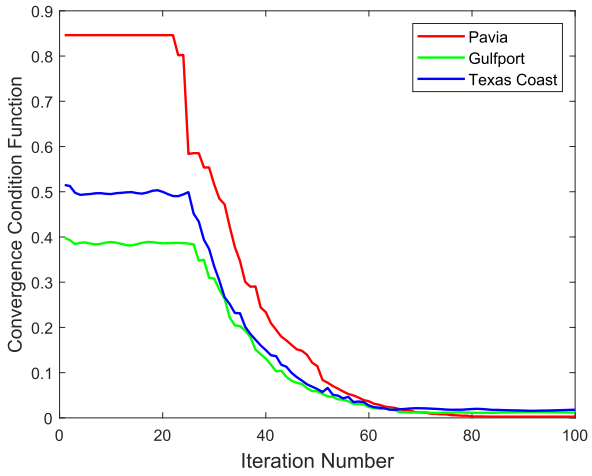


Fig. 11. Change in the convergence condition function versus the iteration number on the Pavia, Gulfport, and Texas Coast datasets.

F. Complexity and Convergence Analysis

1) Complexity Analysis: In this section, we briefly introduce the complexity of the proposed TLRA-MSL method. For the input HSI tensor $\mathcal{H} \in \mathbb{R}^{w\times h\times b}$ after orthogonal transformation, the main computation costs are from updating $\mathcal{C}_s$ and $\mathcal{R}$. The cost of updating $\mathcal{C}_s$ is composed of the computational complexity of t-SVD and FFT, which cost $\mathcal{O}(Kr_{\mathcal{H}}^2 h\lceil(b+1)/2\rceil)$ and $\mathcal{O}(Kr_{\mathcal{H}}hb\log(b))$. K represents the iterations number of the proposed method in Algorithm 3. Meanwhile, the computation cost for updating $\mathcal{R}$ requires $\mathcal{O}(r_{\mathcal{H}}(w+h)b\log(b)+r_{\mathcal{H}}whb)$. Finally, the total computational complexity of the proposed TLRA-MSL is the sum total of these three parts.

2) Convergence Analysis: As shown in Fig. 11, we investigate the astringency of the proposed TLRA-MSL on the Pavia, Gulfport, and Texas Coast datasets. The convergence condition function denotes the condition function in Algorithm 3. From Fig. 11, we can observe that the values of the convergence condition function will gradually converge to 0 when iterations approach the max iteration number. Therefore, the proposed method has good convergence on the HAD task.

V. Conclusion

In this article, we design a TLRA-MSL model to detect abnormal targets in HSI datasets. Our method preserves the inherent 3-D structure of the HSI tensor by direct TLRA. Considering the multisubspace structure in heterogeneous HSI scenes, we incorporate the multisubspace learning strategy to obtain the tensor linear representation of backgrounds. In addition, we apply the ε-TNN to estimate the low-rank component for better background recovery. The $l_{2,1}$ constraint is introduced to the proposed TLRA-MSL model to highlight the group sparsity of anomalous targets. Meanwhile, we design an efficient ADMM algorithm to solve the TLRA-MSL model. Extensive experimental results on six HSI datasets prove the effectiveness and superiority compared with other anomaly detection algorithms with excellent performance.

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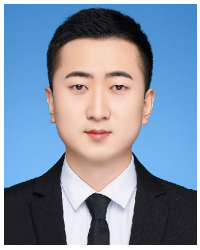
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